

SOFTWARE TESTING PLAN

Graduation Project — Web App & API

Project Name:

Automation Exercise — E-Commerce Web App
Version 1.0
Prepared by: Testing Team (5 Members)

1. Project Overview

Field	Details
Project Name	Automation Exercise
Version	1.0
Application Type	Web Application + REST API
Website URL	https://automationexercise.com
Team Size	5 Members
Description	A full-fledged e-commerce clothing store used for automation practice. Supports user registration, login, product browsing, cart, checkout, and API testing.

2. Testing Scope

The following modules and features will be tested:

- User Registration — sign-up form, fields validation, email validation
- User Login / Logout — correct & incorrect credentials, session management
- Account Deletion — delete flow and confirmations
- Home Page — navigation, recommended items, scroll, footer
- Products — View all, product detail, search, category & brand filters
- Product Review — submit review on product page

- Shopping Cart — add, view, update quantity, remove, totals
- Checkout — 3 order scenarios (register during/before/login before checkout), address & invoice
- Cart Persistence after Login — cart data retained post login
- Contact Us — form submission, file upload, success message
- Subscription — subscribe from home page and footer
- API Endpoints — all REST APIs (productsList, brandsList, searchProduct, verifyLogin, createAccount, getUserDetailByEmail, etc.)

3. Team Roles & Responsibilities

#	Name	Assigned Features
1	Hanan Mohammed Elhady	User Registration, Products (Category & Brand), Scroll
2	Ibrahim Akram Ibrahim	User Login, Products (View/Detail/Search), Product Review
3	Esraa Mostafa Maher	Shopping Cart, Contact Us, Subscription
4	Amira Ali Mohamed	Checkout (3 Order Scenarios), Address & Invoice, Cart Persistence
5	Aya Mohammed	Logout, Account Deletion, Home Page, Subscription

4. Testing Types & Tools

Testing Type	Tool / Framework	Description
Manual Testing	Excel sheet	Hand-executed test cases for UI, login, forms
Automation Testing	Java, OOP, selenium web driver, testNG, Page object model(POM), Data driver testing(DDT), Cucumber(BDD)	Automated regression and smoke tests
API Testing	postman , excel sheet , javaScript	Validate all REST endpoints, status codes, responses
Regression Testing	Java, OOP, selenium web driver, testNG, Page object model(POM), Data driver	Re-run tests after every code change

	testing(DDT), Cucumber(BDD)	
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5. Key Performance Indicators (KPIs)

KPI	Target	Measurement Method
Bug Detection Rate	≥ 90%	Bugs found / total bugs × 100
Test Coverage	≥ 80%	Lines/functions covered by tests
API Response Time	< 2 seconds	Average response time per endpoint
Test Pass Rate	≥ 95%	Passed tests / total executed × 100
Critical Bug Fix Time	< 48 hours	Time from report to resolved

6. Risks & Mitigation

Feature	Risk	Likelihood	Mitigation Plan
User Registration	Email validation fails on edge cases	Medium	Test valid, invalid, duplicate emails
User Login	Session not cleared after logout	Medium	Verify session token on each logout test
Products / Search	Search returns wrong results	Low	Test with exact name, partial, and invalid input
Shopping Cart	Quantity/total calculation errors	Medium	Verify totals after each add/remove action
Checkout	Address not saved correctly	Medium	Compare address at registration vs. invoice
Cart Persistence	Cart lost after	High	Add items as guest, login, verify

	login		cart intact
Contact Us / File Upload	File type/size not validated	Low	Test with valid and invalid file formats
API Endpoints	Wrong status codes or response body	Medium	Validate response schema & status for all APIs

7. Approval & Sign-off

Name		Role	Signature
Abdelrahman Osama		Project Supervisor	